NewsBot Intelligence System - Midterm Individual Reflective Journal

Course: ITAI 2373 | Natural Language Processing

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Project: NewsBot Intelligence System

Project Type: Individual Implementation

GitHub: https://github.com/iamhuongthn/ITAI-2373-NLP-NewsBot-Intelligence-System

1. Project Ownership and Independent Development

I have been working independently on the NewsBot Intelligence System, meaning I am completely responsible for everything from the preparation of the dataset used for training to the implementation of NLP, the evaluation of the trained models, the integration of all modules into a whole system, the testing of all parts, and the documentation of the whole project.

What I mainly wanted to explore, rather than building another news classifier, is how all these different techniques can be combined to generate meaningful and interpretable information from unstructured news articles.

This work was conducted alone. Thus, I was left to make all technical decisions, work through integration issues, and verify whether the output from the system supported meaningful conclusions. In terms of constraints, there were eight different NLP modules that needed to function in concert, while also avoiding overloading the computational resources that Google Colab affords.

The final system consists of eight different NLP modules: text preprocessing, TF-IDF, part-of-speech analysis, syntax parsing, sentiment and emotion analysis, multi-class classification, and named entity recognition.

2. Technical Integration and Challenges

The most important technical challenge was connecting eight NLP modules into one consistent system instead of treating them as separate exercises.

I used the BBC News Classification Dataset, starting with 1,490 records. After removing 50 exact duplicates, the final dataset contained 1,440 articles across five categories: Business, Entertainment, Politics, Sport, and Tech.

For classification, I used a stratified 80/20 split, with 1,152 training articles and 288 testing articles. I also used five-fold cross-validation, fitting preprocessing transformations within each training fold to prevent feature leakage.

Three model configurations were evaluated:

Model	Test Accuracy	Macro F1
Naive Bayes	94.44%	94.03%
Logistic Regression	92.71%	92.36%
SVM	95.83%	95.57%

One interesting decision was selecting Naive Bayes even though SVM achieved higher test accuracy. The best training cross-validation macro F1 score was achieved by the Naive Bayes model, with a score of 97.01%, compared to 95.83% for SVM. I used cross-validation to select the model instead of choosing the highest-scoring model on the held-out test data.

This reinforces an important lesson: model selection is fundamentally different from model evaluation. Even a higher score on the test data is not sufficient to override one's selection procedure based on cross-validation of the training data.

The scores were obtained from complete pipelines with different feature configurations, so the results should be viewed as comparisons of complete systems rather than simply comparisons of algorithms.

3. Analytical Findings and Interpretation

Each of the many different NLP techniques attempts to answer different questions when applied to an article.

I found that using TF-IDF to create features for the classification of news articles helped me understand the category-specific vocabulary used in each article, such as "Business" articles having lots of instances of the words "firm", "company", and "market" compared to "Politics" articles having lots of instances of the words "labour", "election", and "government".

POS and syntax analysis revealed further aspects of the news articles. In the saved analysis for this task, Entertainment had the highest noun-related pattern, Politics the highest verb-related pattern, and Business the highest numerical pattern. Politics also had the longest sentences on average. For the dependency-distance pattern under the implemented analysis, Sport had the highest score.

Sentiment analysis for Politics was clearly the lowest, at about 0.054. For Entertainment, it was the highest, with an average VADER compound score of about 0.590.

Note, though, that the results from the VADER sentiment analysis should not be used as evidence to argue that political reporting is biased or that entertainment reporting is positive.

The NRC experiment was set up to analyze eight different emotion associations in a total of 150 articles, 30 per category. Note, though, that these results should not be confused with human judgments of the emotions expressed in these articles.

NLP output is very powerful and will become even more useful when analyzed in unison while recognizing the limitations of each NLP technique.

4. Named Entity Recognition and System Reliability

I really enjoyed seeing how Named Entity Recognition was implemented and just how different extracting information is from correctly extracting information.

The system extracted 40,803 entity mentions, spread across 18 entity types. It was also able to generate a co-occurrence network consisting of 30 nodes and 222 edges for these entities.

However, the manual NER review revealed errors that the automated summary statistics alone could not identify. The automatically detected name "William Hill", for example, was classified as a person and not as an organization. Another error consisted of including the score of a tennis match within the name of a player.

Thousands of entities can be extracted from text by a system, and yet many mistakes can still be made regarding the correct boundaries of entities as well as their classification. This shows why manual review remains important.

My initial insight here was that just because two entities appear in the same article, it does not necessarily mean that they are related in real life. The co-occurrence network is useful for exploration, but it does not prove real-world relationships.

For end-to-end testing of NewsBot, nine test articles were successfully processed. The articles consisted of seven synthetic examples and two external articles that were manually assigned to categories. Both external articles matched their manually assigned categories.

The results from two external articles are promising but not yet sufficient to adequately test the performance of the system on current news from a variety of publishers.

5. Business Value and Future Improvements

My intention is to use NewsBot as a foundation for a practical news intelligence application that goes beyond the scope of a simple university classification task.

For media organizations, the system could be used for automatically categorizing articles, for entity-based search, for monitoring topics over time, and for analyzing language use in different news categories.

Combining these outputs is more useful for businesses: the category output from the classifier tells the user which topic the story is about; the sentiment score is a measure of the "tone" of the words used in the article; and the entities identified in the article point to people, organizations, and locations.

The prototype presented here has several limitations: there are no original publication dates or independent headlines for the articles in the BBC dataset used. Historical news might not look the same as current news anymore. In addition, the classification probabilities have not been calibrated.

Next, I would like to test the system using a set of independently labeled external test articles, with a particular focus on improving entity recognition, conducting a more systematic feature comparison, and most of all, exploring articles interactively.

Should additional computational resources be available to support even more complex models, an evaluation of their increased accuracy over the simpler models and whether that improvement justifies the additional computational cost would be of interest.

6. Personal Learning and Professional Development

I gained further insight into how to approach NLP as an engineering project in addition to merely applying many different NLP algorithms to a problem.

By completing this project, I was able to develop, integrate, and test many different text-processing techniques. Also, I was able to set up a complete, reproducible test suite, learn how to properly evaluate the results of a classification model, learn how to better inspect errors in text extraction, and even map the results of this project to many potential applications in business.

Separating the model's output from the evidence supporting said output was also particularly instructive.

A 94.44% classification accuracy on the BBC dataset used for evaluation does not necessarily translate to the same high level of performance on other sources of news. In addition, while the sentiment and entity analyses revealed some very useful patterns, these do not necessarily prove bias, emotion, or real-world relationships.

As I completed this project independently, I have also had the opportunity to take ownership of the full development lifecycle of a project, including all of the technical decisions as well as their limitations.

A useful AI system does not only have to predict something. More importantly, it has to provide information that can be interpreted, evaluated, and used in a responsible manner by humans. NewsBot helped me move from individual NLP methods to this perspective.